

SUMMARY

Data Science Master’s student proficient in Python, PyTorch, and TensorFlow. Experienced in solving complex problems across various domains, including predictive modeling, computer vision, and Generative AI. Skilled in data engineering and model optimization, with the ability to bridge the gap between data science and web application deployment using tools like React, Docker, and Streamlit.

TECHNICAL SKILLS

- Languages: Python, Java, JavaScript, TypeScript
- AI/ML: TensorFlow, Keras, scikit-learn, Pandas, NumPy, PyTorch, NLP (NLTK, Spacy, Transformers), Deep learning, Speech Processing, CNN, RAG, LLM, SLM
- Web: React.js, Next.js, Node.js, REST APIs, Django, Flask, FastAPI
- Databases: MongoDB, MySQL, ChromaDB
- Tools: Git, GitHub, Docker, Streamlit, Jupyter Notebook, LangChain, Ollama, Hugging Face

PROJECTS

- Automated Fact-Checking Pipeline via LLM Fine-Tuning** : I engineered a synthetic training dataset via knowledge distillation (using Gemini 2.5 Pro) and fine-tuned the Mistral-Nemo 12B model using Unsloth (QLoRA) to achieve a 0.92 training loss. I also designed a RAG retrieval pipeline using LangChain and ChromaDB to ground the model’s outputs in verified data from Politifact and Snopes, ensuring zero hallucinations on extraction tasks.
- Brain Tumor Detection Using Deep Learning** : I developed a deep learning model for brain tumor detection, utilizing a VGG16 architecture with transfer learning to accurately classify MRI images into four categories: glioma, meningioma, pituitary, or no tumor, using Python, TensorFlow, and Keras.
- Movie Recommendation System** : I created a content-based movie recommendation system using a KNN algorithm with scikit-learn and deployed it as an interactive web application using Streamlit.
- House Price Prediction Using Linear Regression** : I built a linear regression model that involved extensive data preprocessing and feature engineering, which achieved strong performance metrics such as an R-squared value of 0.8469 .

EXPERIENCE

Integral Progress Technology <i>Front-End Developer</i>	Marrakech, Morocco <i>Apr 2024 - Aug 2024</i>
- Developed dynamic web pages for a Quality Management module using React.js, TypeScript, and Tailwind CSS. - Integrated APIs to fetch and display dynamic data , Managing application state using Redux Toolkit. - Collaborated with backend team on API communication and error handling.	

EDUCATION

Polydisciplinary Faculty of Safi - Cadi Ayyad University <i>Master’s in Data Science and Analytics</i>	Safi, Morocco <i>Oct 2024 - July 2026</i>
Essaouira Higher School of Technology - Cadi Ayyad University <i>Professional Bachelor’s Degree in Computer Systems and Software Engineering</i>	Essaouira, Morocco <i>Oct 2023 - July 2024</i>

LANGUAGES

Arabic (Native)	French (Professional)	English (Professional)
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